Do Now · retrieval

Trees and binary search trees with arrays

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an array, and how do you access an individual element?

Q2

What is the purpose of a pointer in a linked list?

Q3

Explain the difference between LIFO and FIFO. Name one data structure using each.

Q4

A graph has vertices A, B, C and D. Its undirected edges are A–B, A–C and C–D. Write its adjacency list.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

BST ordering — CODE:
smaller → left
larger → right — Each comparison chooses the next branch.

Core idea 2

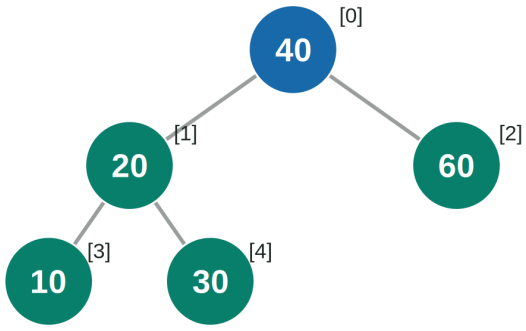
Search — CODE:
current = root;

while (current != -1 && data[current] != target)
{
if (target < data[current])
current = left[current];
else
current = right[current];
} — Search follows only one branch at each comparison.

Core idea 3

Add — CODE:
if (value < data[parent])
left[parent] = newIndex;
else
right[parent] = newIndex; — The new index is attached to the correct null pointer.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Remove simple cases — CODE:
leaf → parent pointer = -1

one child → parent pointer = child index — Two-child removal is more involved; first secure leaf and one-child cases.

Vocabulary · use the correct technical terms

tree — A hierarchical structure of nodes.

root — The top node.

parent — A node with child nodes.

child — A node directly below a parent.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Search for 30

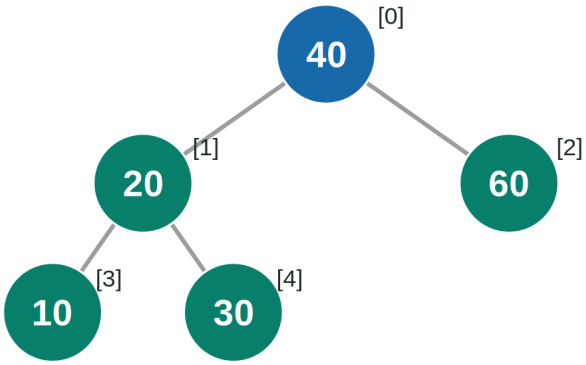
Follow the worked method

1. At index 0 — CODE:
30 < 40
current = left[0]
current = 1

2. At index 4 — CODE:
data[4] = 30
found

3. Skipped nodes — CODE:
Did not inspect indexes 2 or 3.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete one row
CODE:
index 0: Sligo
left = ?
right = ?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Build a BST with parallel arrays
Represent 40 with children 20 and 60, then 20 with children 10 and 30.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Remove a leaf using a pointer change

Remove node 10 at index 3 from the example BST, then search for 10.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In an array-based BST, left and right fields store indexes/pointers, not copies of the child data.

Use key vocabulary from memory

1. Explain tree in your own words.

2. Explain root in your own words.

3. Explain parent in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.